SECP2523

DATABASE (WBL)

SECTION 02

ASSIGNMENT 1

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QUESTION 1

One of the application areas of database systems is e-commerce such as online shopping.

What are the data or information stored in the database system of an online shopping website? Give FOUR (4) of them.

Data	1. Order 2. Payment 3. Product 4. Customer
Information	1. Iphone 16 2. Online Banking 3. Casing Iphone 16 4. Address

QUESTION 2

Answer the questions below based on the following relational schema:

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SCIENTIST (Sc_ID, Sc_Name, Sc_Phone, Sc_Add)

EXPERIMENT (Exp_NO, Ex_Name, Proj_Code)

PROJECT (Proj_Code, Proj_Name, Proj_StartDate)

SCIENTIST_EXPERIMENT (Sc_ID, Exp_NO, Date, Time)
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a) For each table, identify the primary key and foreign key. If a table does not have a primary key or foreign key, write – in the space provided. (4M)

TABLE	PRIMARY KEY	FOREIGN KEY
SCIENTIST	Sc_ID	-
EXPERIMENT	Exp_NO	Proj_Code
PROJECT	Proj_Code	-
SCIENTIST_EXPERIMENT	Time	Sc_ID, Exp_NO

b) Does the table exhibit entity integrity? Answer **'Yes'** or **'No'** and then explain your answer. (4M)

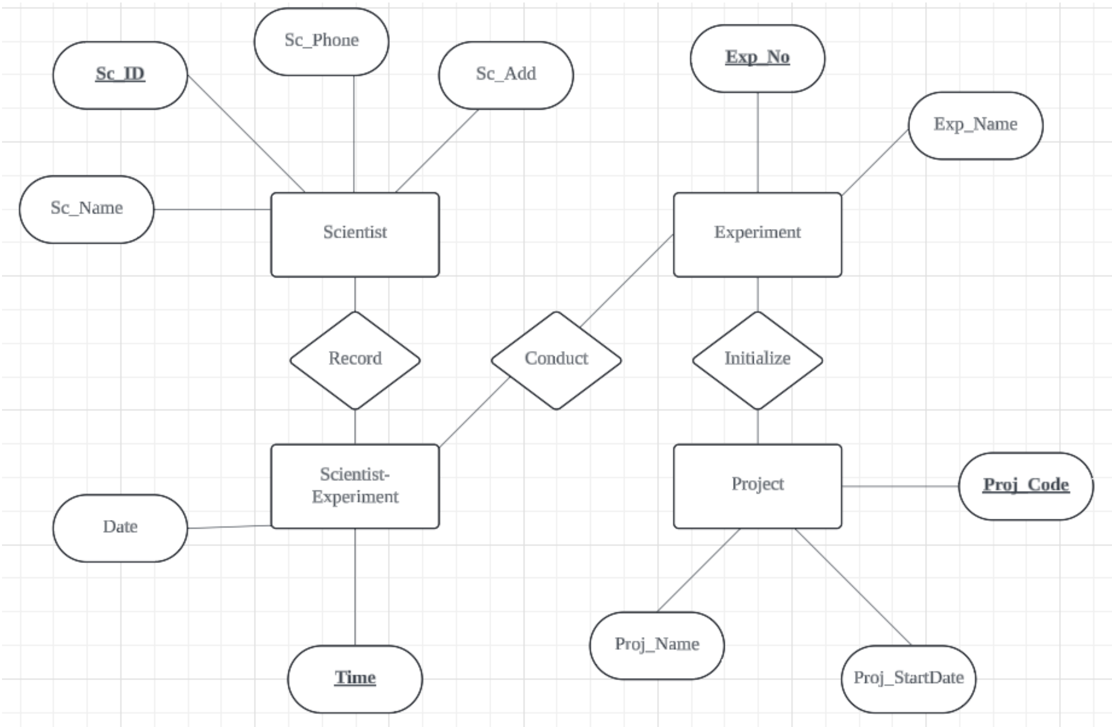
TABLE	ENTITY INTEGRITY	EXPLANATION
SCIENTIST	Yes	Because it has a primary key.
EXPERIMENT	Yes	Because it has a primary key.

PROJECT	Yes	Because it has a primary key.
SCIENTIST_EXPERIMENT	Yes	Because it has a primary key.

c) Does the table exhibit referential integrity? Answer **'Yes'** or **'No'** and then explain your answer. (4M)

TABLE	REFERENTIAL INTEGRITY	EXPLANATION
SCIENTIST	No	Because it has no foreign key.
EXPERIMENT	Yes	Because it has a foreign key.
PROJECT	No	Because it has no foreign key.
SCIENTIST_EXPERIMENT	Yes	Because it has a foreign key.

d) Draw ERD using Chen notation based on the given relational schema. (4M)



QUESTION 3

Read the following and complete the task listed.

Consider the following about Luxury Records:

Here is the information that you gather:

- Luxury has decided to store information about musicians who perform on its albums (as well as other company data) in a database.
- Each musician that records at Luxury has an SSN, a name, an address, and a phone number.
- Poorly paid musicians often share the same address, and no address has more than one phone.
- Each instrument used in songs recorded at Luxury has a unique identification number, a name (e.g guitar, synthesizer, and a flute) and a musical key (e.g., C, B-flat, E-flat).
- Each album recorded on the Luxury label has a unique identification number, a title, a copyright date, a format (e.g., CD or MC) and an album identifier.
- Each song recorded at Luxury has a title and an author.
- Each musician may play several instruments, and a given instrument may be played by several musicians.
- Each song album has a number of songs on it, but no song may appear on more than one album.
- Each song is performed by one or more musicians, and a musician may perform a number of songs.
- Each album has exactly one musician who acts as its producer.
- A musician may produce several albums, of course.

Task:

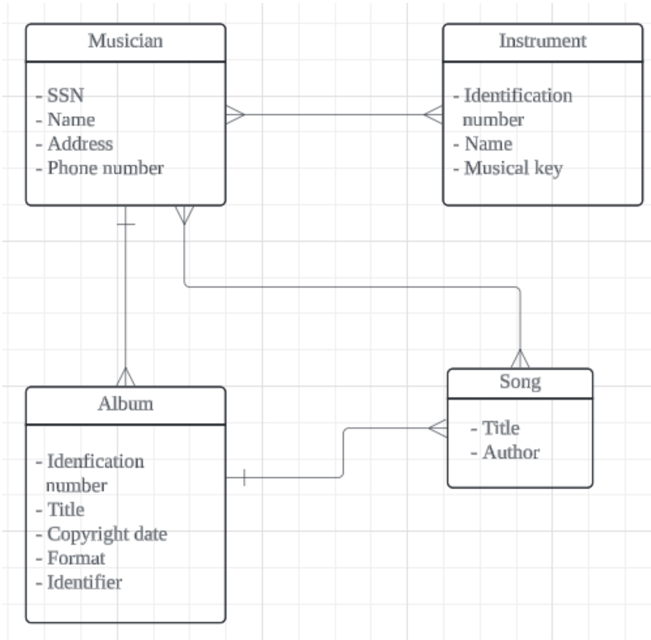
1. List each Entity that is identified in Luxury Records System.
2. For each entity list the attributes.
3. Draw a logical model to represent each entity and the relationships between them.
4. Draw an Entity Relationship Diagram (ERD) that captures the entire system.

Ans:

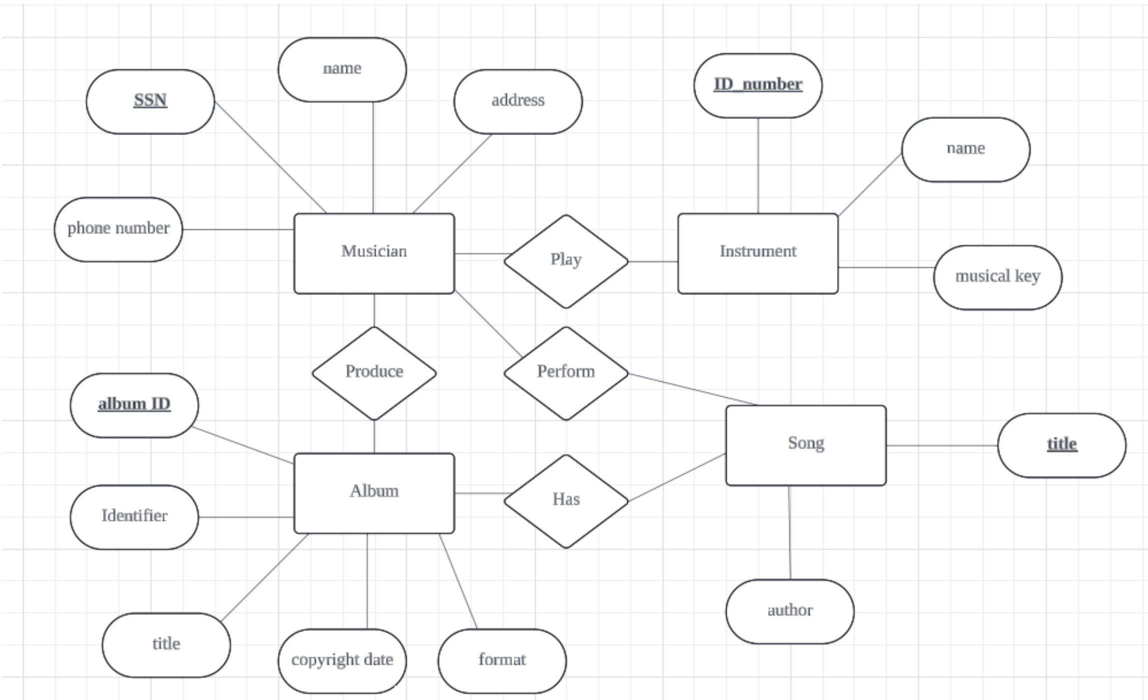
1, 2.

Entity	Attribute
Musician	SSN, name, address, phone number
Instrument	ID number, name, musical key
Album	ID number, title, copyright date, format, album ID
Song	Title, author

3.


Musician
- SSN
- Name
- Address
- Phone number
Instrument
- Identification number
- Name
- Musical key
Album
- Idenfication number
- Title
- Copyright date
- Format
- Identifier
Song
- Title
- Author

4.


SSN
name
address
ID_number
name
phone number
Musician
Play
Instrument
musical key
Produce
Perform
album ID
Song
title
Album
Has
Identifier
author
title
copyright date
format

QUESTION 4

Give an example of each of the three types of relationships.

i) One-to-many relationship

- A customer can buy many products.

ii) Many-to-many relationship

- A musician can play many song, a song can be played by many musicians.

iii) One-to-one relationship

- A student can only take a program.

iv) Many-to-one relationship

- Multiple songs can have only one producer.

QUESTION 5

CLIENT (Client_Number, Clent_Name, Street, City, State, Postal_Code, Amount_Paid, Current_Due, Recruiter_Number)
RECRUITER (Recruiter_Number, Last_Name, First_Name, Street, City, State, Postal_Code, Rate, Commission)

a) What is the relationship between Recruiter and Client?

- Many to one

b) What is the primary key for the Client table and Recruiter table?

- Client_Number
- Recruiter_Number

c) Define foreign key? Identify foreign key in the Client table.

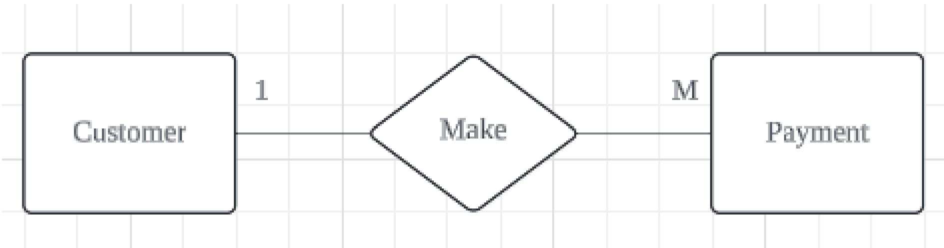
- An attribute of an entity which serve for another entity. Recruiter_Number.

QUESTION 6

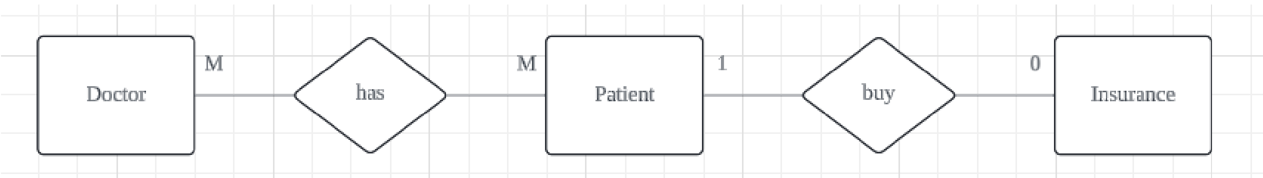
Draw the Chen & Crow's Foot notation based on the following statements:

i) "A customer can make many payments, but each payment is made by only one customer."
ii) "A medical doctor may treat many patients. Some patients may be treated by more than one medical doctor. A patient may have or may not have insurance plan."

i)



ii)



QUESTION 7

Consider the following relations for a database that keeps track of student enrollment in courses and the books adopted for each course:

`STUDENT(SSN, Name, Major, Bdate)`

`COURSE(Course#, Cname, Dept)`

`ENROLL(SSN, Course#, Quarter, Grade)`

`BOOK_ADOPTION(Course#, Quarter, Book_ISBN)`

`TEXT(Book_ISBN, Book_Title, Publisher, Author)`

Draw a relational schema diagram specifying the foreign keys for this schema.